



Power Consumption Forecasting Report

Project Title:

Predictive Analytics for Power Consumption in Zone 1 – Tetouan City



Business Objective

To empower energy stakeholders with data-driven insights for proactive energy management in **Zone 1 of Tetouan City**.

The predictive model supports:

- Smarter resource planning
 - Demand forecasting
 - Operational efficiency
 - Informed decision-making for energy distribution
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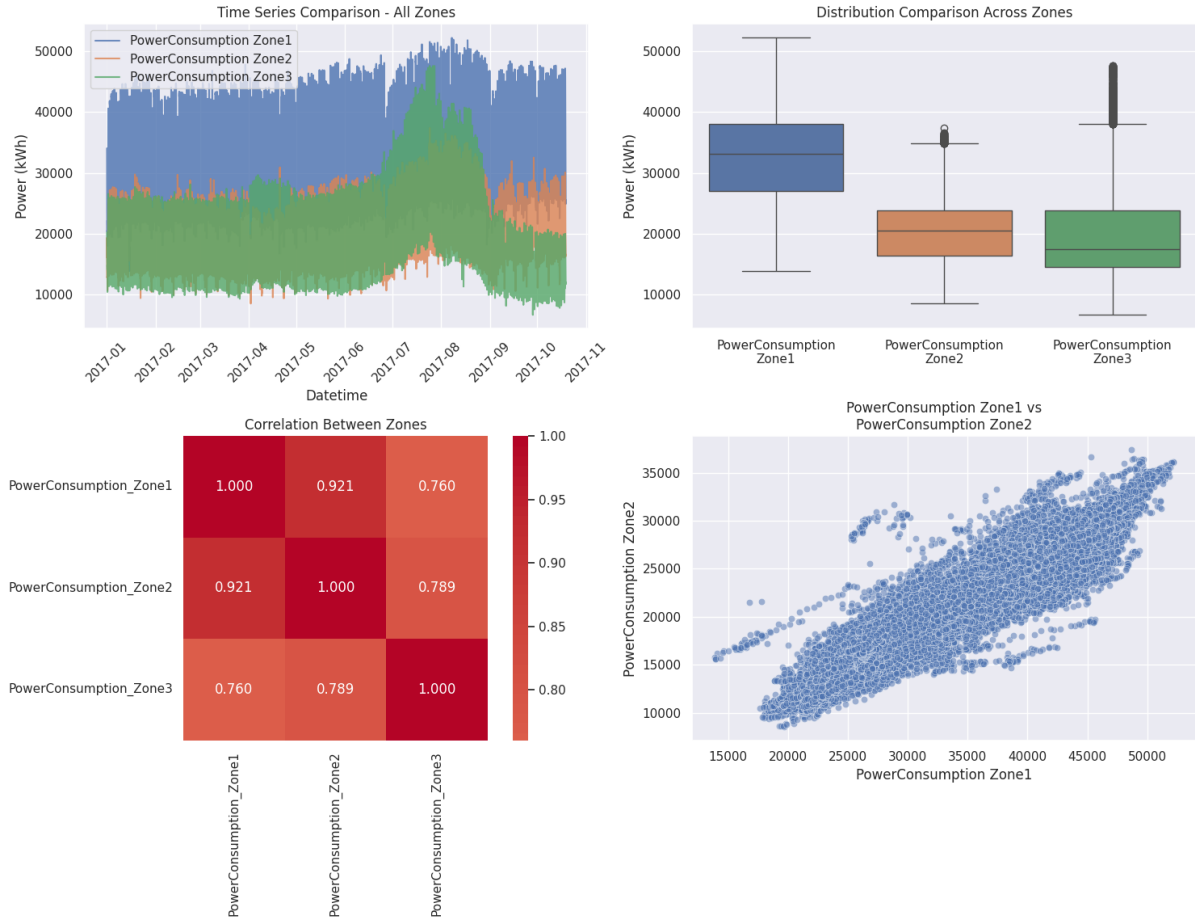
Data Overview

- **Source:** UCI Machine Learning Repository
 - **Collection Frequency:** Every 10 minutes
 - **Key Variables:**
 - Environmental Factors: Temperature, Humidity, Wind Speed, Diffuse Flows
 - Power Consumption: Zone 1 (Forecast Target), Zone 2, Zone 3 (Reference Data)
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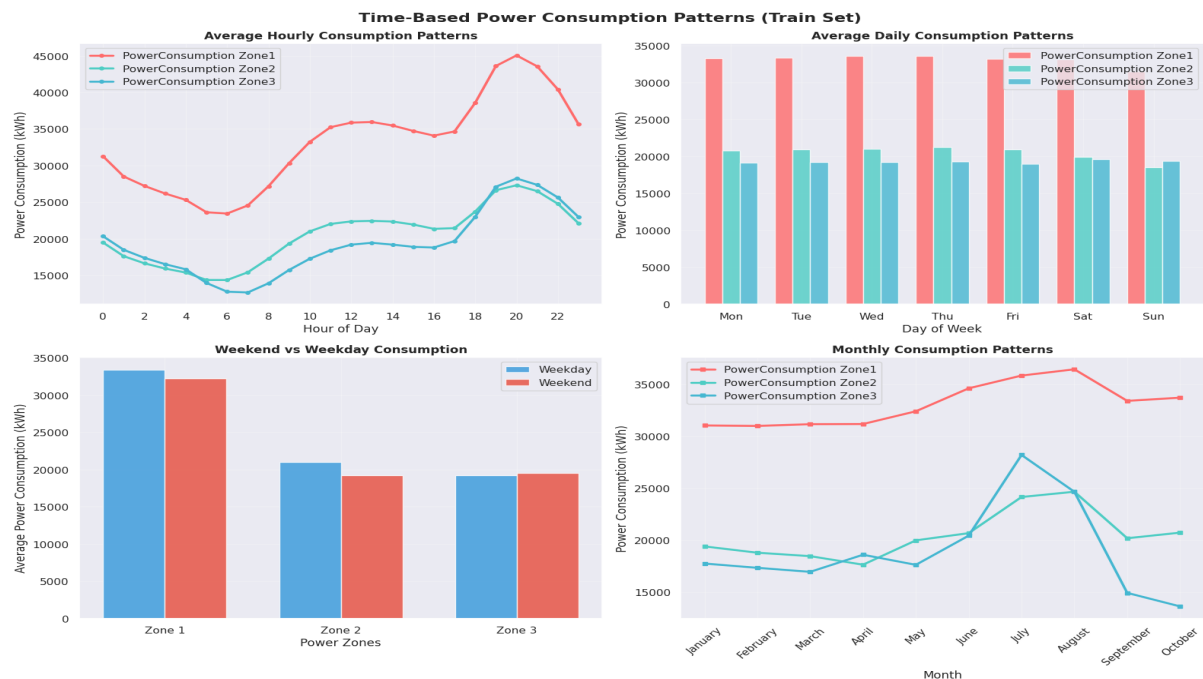
🌱 Key Business Insights

- **Stable Power Demand in Zone 1:**
Continuous, high usage suggests critical operational demand in this area.

Multi-Zone Power Consumption Analysis (Train Set)



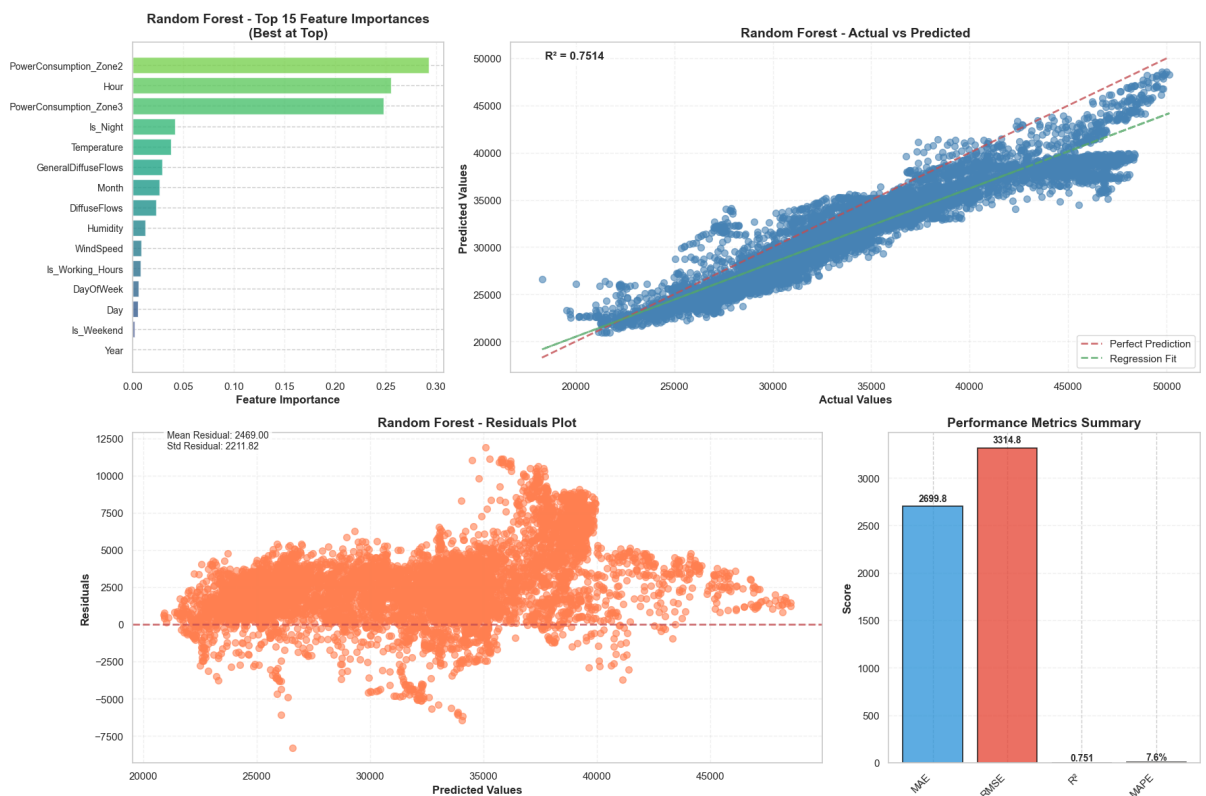
- **Seasonality & Weather Impact:**
Power usage patterns correlate with temperature and seasonal changes, highlighting the need for adaptive demand strategies.



- **Cross-Zone Influence:**
Zone 2 and Zone 3 consumption strongly relate to Zone 1 demand—indicating shared operational drivers.
- **Significant Environmental Predictors:**
 - **Humidity & Diffuse Flows:** Increase in these factors leads to higher power usage.
 - **Wind Speed:** Acts as a demand reducer.
 - **Temperature:** Surprisingly, not a significant driver for Zone 1 demand.

Modeling Outcome

- **Model Used:** Random Forest Regressor
- **Performance:** R^2 Score of 0.758 on the test set (explaining 75.8% of variance)



✅ **Business Value of Model:**

- Predicts power consumption with high accuracy
 - Enables better load management and operational forecasting
 - Supports strategic planning for energy provision
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✅ **Recommendations for Stakeholders**

- **Deploy the Model in Operations:**
Use forecasts for daily and weekly energy management.
 - **Integrate with Monitoring Systems:**
Real-time dashboards for proactive insights and alerts.
 - **Consider Model Enhancements:**
Test advanced ensemble methods (e.g. LSTM) for further optimization.
 - **Maintain and Update Models Regularly:**
Update forecasts with new data to ensure ongoing relevance.
 - **Extend Predictive Approach City-Wide:**
Apply this methodology to other city zones for comprehensive energy planning.
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🌟 **Business Impact Summary**

By leveraging predictive analytics:

- ✅ **Operational Efficiency Improves**
- ✅ **Costs are Reduced through Smarter Planning**
- ✅ **Energy Sustainability Goals are Supported**

Data-driven forecasting ensures Tetouan City is equipped for smarter, future-ready energy management.